$\delta \cdot \phi \to \phi(0)$

输入一个函数，输出一个数 $\phi(0)$

对于delta，它只用一个值，没有其他的

Math 534H Lec18

$$u(x,t) = \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{\infty} e^{-(x-y)^2/4t} \phi(y)\, dy$$

$$u(x,0) = \phi(x)$$

$$\lim_{t\to 0^+} u(x,t) = \phi(x), \text{ Have?}$$

相当理论于原点的Dirac delta

From lec17. $\lim_{t\to 0^+} \underbrace{S(x,t)}_{\text{Heat Kernel}} = \delta_0 / \delta(x)$ Dirac delta at the origin

What this means if ϕ also is smooth at 光滑函数

$$\delta_0 \cdot \phi \to \phi(0)$$

δ_{x_0} : Dirac$_{x_0}$ delta on point x_0

This means

$$\delta_{x_0} : \phi \to \phi(x_0)$$

Then when you take $\lim_{t\to 0} u(x,t)$ you get this limit equals

"$\int \delta(x-y)\phi(y)\,dy$"

But this is not an integral only more but is a pairing of (Distribution)

In other words

$$\int \delta(x-y)\phi(y) = \boxed{\phi(x)}$$

当作$\delta(x-y)$对应到函数$\phi(y)$的分布

Bcuz this simply means that you remember Dirac Delta at

$$\int \delta(x-y)\phi(y)\,dy = \int \delta_x(y)\phi(y)\,dy = \phi(x)$$

$$\delta_{x_0}(\phi) = \phi(x_0) \;/\; \langle \delta, \phi \rangle = \phi(0) / \delta(0) = \phi(0)$$

Uniqueness of solution to heat equation on R

Problem 9(2.4)

$$\begin{cases} u_t - k u_{xx} = 0 \\ u(x,0) = x^2 \end{cases} \quad x \in R,\ k>0,\ t>0$$

1) note u_{xxx} also solve heat equation bec it's initial condition equal to 0.

2) Integrating 3 times, from 1) you get that the general form for

$$u(x,t) = A(t)x^2 + B(t)x + C(t)$$

3) Take formula in 2) you plug in $u_t - k u_{xx} = 0$

$\Downarrow$

Get relation of $A(t), B(t), C(t)$

$A(t) = A_0$ constant in t

$B(t) = B_0$ constant in t

$C(t) = 2kA_0 t + C_0$ C_0 constant

$A'(t) = 0$

$B'(t) = 0$

$C'(t) = 2kA(t)$

4) Finally, use initial condition $u(x,0) = x^2$

$A_0 = 1,\ B_0 = 0,\ C_0 = 0.$

5) Get

$$u(x,t) = x^2 + 2kt$$

What happen if we use heat kernel representations.

$$u(x,t) = \frac{1}{\sqrt{4k\pi t}} \int_{-\infty}^{\infty} e^{-(x-y)^2/4kt} \phi(y)\,dy$$

$$\phi(y) = y^2$$

$$\boxed{-(\sqrt{4kt})p + x}$$

y^2

$$p = \frac{x-y}{\sqrt{4kt}} \qquad dp = -\frac{1}{\sqrt{4kt}}\,dy$$

$$dy = -\sqrt{4kt}\,dp$$

$$u(x,t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-p^2}\,dp \cdot (-\sqrt{4kt}) \cdot y^2$$

$$= +\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-p^2} (x - \sqrt{4kt}\,p)^2\,dp$$

① $x^2 \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-p^2}\,dp$

$$= x^2 \cdot \frac{1}{\cancel{\sqrt{\pi}}} \cdot \cancel{\sqrt{\pi}} = x^2$$

② $-\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \sqrt{4kt}\, x p\, e^{-p^2}\,dp$

③ $4kt \underbrace{\int_{-\infty}^{\infty} p^2 e^{-p^2}\,dp}_{= I} \frac{1}{\sqrt{\pi}}$

$$\boxed{u(x,t) = x^2 + \frac{1}{\sqrt{\pi}} \cdot 4kt \cdot I}$$

$$u(x,t) = x^2 + 2kt$$

So $\cancel{2kt} = \frac{2}{\sqrt{\pi}} \cancel{2kt}\, I$

$$1 = \frac{2}{\sqrt{\pi}} I$$

$$I = \frac{\sqrt{\pi}}{2}$$

$$\int_{-\infty}^{\infty} p^2 e^{-p^2}\,dp = \frac{\sqrt{\pi}}{2}$$

Chapter 4

BIVP for wave equation

homogeneous Dirichlet boundary conditions on $[0, L]$

$$\begin{cases} \text{EQ} & u_{tt} - c^2 u_{xx} = 0, \quad x \in [0, L], \quad t > 0 \\ \text{BC} & u(0, t) = 0 = u(L, t), \quad \forall t > 0 \\ \text{IC} & u(x, 0) = \phi(x) \\ & u_t(x, 0) = \psi(x) \end{cases}$$

① First, consider BC

Make a Guess (ansatz) and look for a solution

$$u(x, t) = X(x)\,T(t) \quad \text{seperation of variable}$$

$$X(x)\,T''(t) - c^2 (X''(x) \cdot T(t)) = 0$$

$$\frac{T''(t)}{c^2 T(t)} = \frac{X''(x)}{X(x)} = \text{constant}$$